

NITIK JAIN

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Graduate student at Johns Hopkins specializing in Embodied AI and Robotics. Previously, spent over 3 years shipping production AI systems across high-impact domains including healthcare AI, autonomous systems, and fintech.

Seeking **Robotics AI, Computer Vision, Perception, and Applied ML** roles.

EDUCATION

Johns Hopkins University

Aug 2025 – May 2027

MSE Robotics (Perception & Cognitive Systems)

Baltimore, MD

- J. N. Tata Scholar & K. C. Mahindra Scholar 2025 amongst ~100 scholars selected worldwide for postgraduate studies

Indian Institute of Technology (IIT), Kanpur

Jul 2017 – May 2022

Integrated B.Tech + M.Tech, Mechanical & Aerospace Engineering · Minor in Machine Learning · GPA 9.4/10

Kanpur, India

- Academic Excellence Award 2021 — top academic performance across the dual-degree program

PUBLICATIONS & PATENTS

- Multimodal Object Detection Under Sparse Forest-Canopy Occlusion  Preprint (extends M.Tech thesis), 2026
- Computer-Implemented Stigma Separation System  Indian Patent No. 564565, 2025
- IR Models and the COVID-19 Pandemic: A Comparative Study  Europe PMC, 2023
- Development of a Low-Cost Autonomous Ground Vehicle  IEEE ICARSC, 2022
- Lightweight Multi-Drone Detection and 3D-Localization via YOLO  arXiv, 2022

WORK EXPERIENCE

Lead AI Engineer, Swiffylabs.ai

Jun 2024 – Jun 2025

WestBridge Capital-backed B2B fintech; owned AI product + engineering for a 0→1 core-banking platform

Bangalore, India

- Led AI product and engineering for a 0→1 core-banking platform, translating NBFC workflows into modular services
- Shipped LLM-agent and fraud-detection automation across onboarding, transactions, and customer-experience workflows
- Drove commercial go-live of a regulatory-ready banking suite across **3 international markets**, launching in **50% less time** than the legacy industry standards

Founding AI Engineer, Osfin.ai

Apr 2023 – Jun 2024

Sequoia-backed fintech building AI-first financial automation

Bangalore, India

- Owned AI product + engineering go-live for a large private bank reconciliation suite processing **4M+ transactions/day**
- Enabled OCR for multilingual unstructured invoice text, improving extraction and matching accuracy by **40%**
- Shipped exception-management workflows that cut recon TAT by **4x**, freed finance-ops teams from manual review, and prevented **400k** in financial leakage at that client

AI Scientist, R&D Division, Qure.ai

Sept 2022 – Mar 2023

3D medical-vision R&D for qCT-Chest, Qure.ai's flagship CT-based lung-health AI suite

Bangalore, India

- Built 3D vision models for CT lung-nodule classification, achieving **88% AUC / 96% recall** on malignancy prediction
- Improved nodule segmentation using attention-augmented U-Net models with CBAM + SE blocks, increasing DICE by **14%**
- Advanced qCT workflows for nodule detection, quantification, and follow-up, supporting deployments processing **1,200 CTs/week** across **4 NHS Trusts**

AI Research Consultant, MooVita Pte Ltd

Sept 2022 – Feb 2023

Computer-vision perception and explainability for autonomous driving

Singapore

- Reconstructed a **3D sparse point cloud** from 2D features on monocular visual odometry from lane detection
- Engineered a **classical optical-flow** fallback for rapid vehicle-turn estimation at **240 FPS** on CPU

RESEARCH EXPERIENCE

Student Researcher, Johns Hopkins Medicine

Aug 2025 – Present

Deep learning for cardiac-MRI analysis Advisor: Dr. Mike Schär

Baltimore, MD

- Developing deep models to **localize coronary arteries in cardiac MRI**, replacing slow, operator-dependent manual scan planning
- Building the data-curation and preprocessing pipeline and evaluating localization accuracy against expert annotations across patients

Sr. Student Researcher, IGCL, IIT Kanpur

Jun 2021 – Aug 2022

M.Tech Thesis: Object Detection Underneath Sparse Occlusion of Forests

Advisor: Dr. Mangal Kothari

Kanpur, India

- Built an **RGB–thermal sensor-fusion** model with **Airborne Optical Sectioning** synthetic-aperture imaging to see through canopy
- Achieved **~70%** occlusion removal and a **32%** detection-accuracy gain by fine-tuning **YOLOv5** on the fused imagery

Visiting Research Intern, RRC, IIIT Hyderabad

May 2019 – Jul 2019

Motion Planning for Outdoor Autonomous UAVs Advisor: Prof. K. Madhava Krishna

Hyderabad, India

- Implemented **VINS-Mono** on ROS, fusing a hardware-synced camera, IMU, and **GPS-RTK** for drift-robust UAV state estimation
- Built a voxel **3D mapper** from RealSense depth, enabling real-time outdoor path planning via `mav_voxblox_planning`

SELECTED PROJECTS

PAVE — Probing Affordance in Vision-Language-Action Encoders

Spring 2026

Do robot-policy (VLA) models keep the affordance signal of their foundation vision encoders?

- A frozen **DINOv2-large** linear probe reaches **0.776 mIoU** on UMD Part-Affordance — **+10.6 pp** over a CVPR'26 dense decoder, with no decoder, fine-tuning, or augmentation
- Showed end-to-end VLA fine-tuning (π_0 , $\pi_{0.5}$, OpenVLA) degrades affordances **class-asymmetrically** (cut -27 pp vs. contain -1 pp), generalizing across two VLA families
- A **297K-param** MLP adapter recovers cut IoU $0.181 \rightarrow 0.405$ (**82%** of the gap); per-class feature drift predicts the loss ($\rho = 0.90$)

Extreme-Illumination Image Restoration

Fall 2025

Restoring detail from low-light and multi-exposure RGB

- Designed a lightweight, **SID-inspired U-Net** trained end-to-end on raw sensor data for low-light / multi-exposure enhancement
- Reached **18.44 dB PSNR / 0.782 SSIM** — **+6.95 dB** over Zero-DCE++ — at a fraction of the parameters

GENIE — Latent Structure with Diffusion Models

Spring 2026

GSoC 2026, ML4SCI — generative modeling of quark/gluon-jet calorimeter images

- Built a VAE $\rightarrow$ **latent-DDPM** pipeline to learn and generate jet images across a 139K-sample calorimeter dataset
- VAE reconstruction **37.93 dB / 0.967 SSIM**; **GraphSAGE** jet classification **ROC-AUC 0.774**; latent DDPM generation at 30.32 dB

Daksh II — Autonomous Ground Vehicle

Aug 2018 – May 2021

Team ViSiON, IIT Kanpur — outdoor self-driving research platform

- Built an end-to-end **ROS autonomy stack**: Fast-SCNN semantic segmentation, LiDAR–camera fusion occupancy mapping, and costmap + RRT path planning
- Led the **21-person** team to **4 international awards** at IGVC 2019, Michigan, incl. the **Lescoe Cup** (2nd worldwide)

AWARDS & ACHIEVEMENTS

- **Lescoe Cup — 2nd Worldwide & 3rd AutoNav**, 27th Intelligent Ground Vehicle Competition, Michigan, USA 2019
- **SIIC Student Innovation Award** (Saffron Stigma Separator); **AWS ML Scholarship**; **NVIDIA Academic Hardware Grant** 2020–22
- **Silver** (TCTD, 7th Inter-IIT) & **Bronze** (ATC, 6th Inter-IIT) medals; **All-India Rank 1116 / 1.5M** in JEE Advanced 2017–19

LEADERSHIP, ENTREPRENEURSHIP & SERVICE

- **Team Leader, Team ViSiON, IIT Kanpur** — led a 21-person team; raised **\$35,000** and an NVIDIA grant; 2nd worldwide at IGVC 2019 2019–21
- **Co-Founder, Tralee.ai (SkipQ)** — AI retail checkout-automation MVP; raised **\$12K** pre-seed; national finalist (top 10 / 1,600+) 2020–21
- **Teaching Assistant**: Innovative Data Strategy for Global Leaders (JHU, 2026) & AE640A Autonomous Systems (IIT Kanpur, 2022)
- **Invited Talk** on ML & Computer Vision (QIP Short-Term Course, IIT Kanpur); **Reviewer**, Seventh Indian Control Conference (ICC-7)

TECHNICAL SKILLS

Languages: Python (*proficient*), C/C++, MATLAB, SQL, Bash, R

ML / DL: PyTorch, PyTorch Lightning + Hydra, MONAI, Hugging Face, LangChain, Scikit-learn, OpenCV, NumPy

Robotics & Tools: ROS + Gazebo, Docker, Databricks, Grafana, CloudWatch, Git, Autodesk Inventor, $\LaTeX$